

CHG 569

Environmental Pollution Engineering

Module 1

(Introduction to the Environmental Problem)



Examining the keywords

- Environment

Atmosphere
Hydrosphere
Lithosphere

} Biosphere — makes life on earth possible

- Pollution — to pollute is to defile, corrupt, contaminate

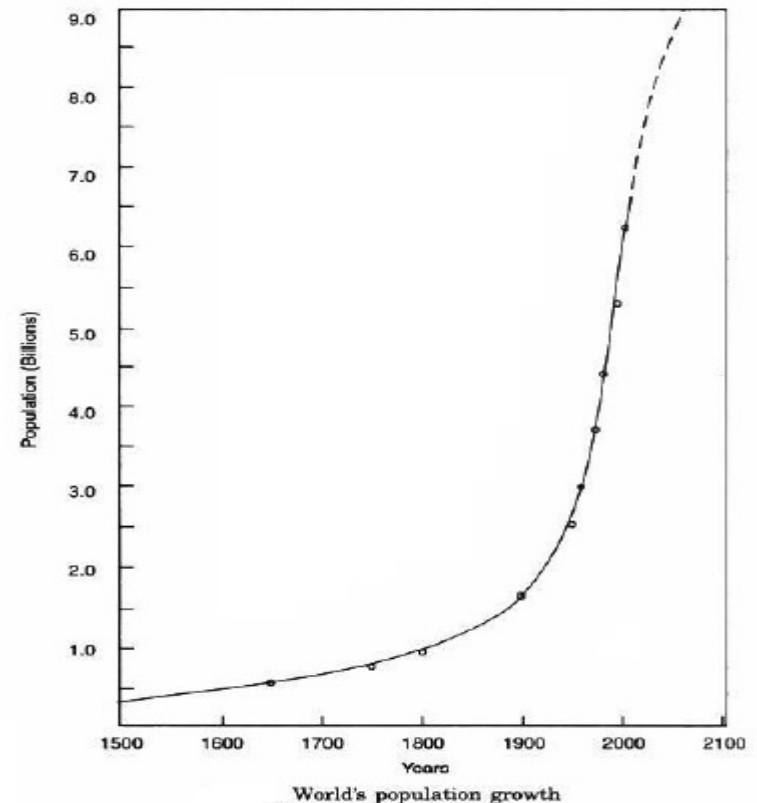
- Engineering — the profession that uses scientific principles to either meet man's need or solve man's problem

Introduction to the Environmental Problem

How did it all start?

- 1 - Human Population growth

Before the industrial revolution, world's population increased in a slow and linear manner. After the eighteen century, growth rate was exponential



Population growth and doubling time

<i>Year</i>	<i>Population (millions)</i>	<i>Doubling time (years)</i>
8000 B.C.	5.32	1000
4000 B.C.	86.5	6400
A.D.O.	133	830
1650	545	240
1750	728	160
1800	906	120
1900	1610	87
1950	2515	
1960	3019	
1970	3698	38
1980	4450	(between 1950-2000)
1990	5292	
2000	6200	
2050 (projected)	9000-11000	

Introduction to the Environmental Problem

How did it all start?

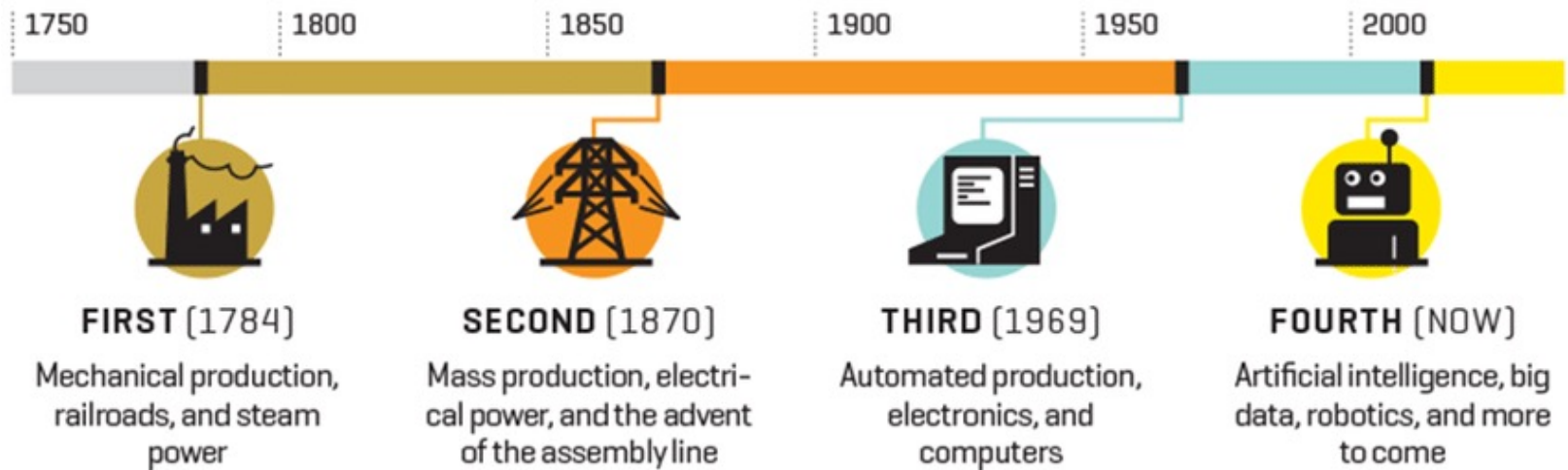
- 2- Industrialization to meet man's needs

So, what really is the problem?

(New chemicals are being developed everyday and these may find their way into the environment thru accidental leakage, during manufacture, transport or usage)

- E.g. - The Bhopal tragedy
- "Erin Brockovich" story
 - Tragedy at Love Canal

The Industrial Revolutions



Industrial Revolution and Society 5.0

1.0 Mechanisation



- The introduction of industrial production equipment driven by water and steam power.

2.0 Electrification



- Mass production using electrical energy and assembly lines.

3.0 Automation



- Automated production due to the rise of electronics, telecommunications and computers.

4.0 Digitalisation



- The use of cyber physical systems on connected devices to automate processes further.

5.0 Personalisation



- The interdependence of man and machine using cognitive computing and human intelligence. Mass customization and personalization for humans.

1780

1870

1970

2011

2020

Society (values & beliefs of people) 5.0 is about integrating AI more effectively into society

Since man appeared on the face of the earth, he has exploited and modified the environment to his advantage in many ways.

Until recently, these changes were only a small perturbation in a large system and had not seriously threatened the homeostatic mechanisms called the feedback loops, which tend to maintain the system in a stable condition.

Unfortunately, now man is capable of causing large enough perturbations in the ecosystem, [The biosphere is made up of so many smaller units aka ecosystems, which in turn can be divided into 2- biotic (living) and abiotic (non-living) components. Abiotic are such things as phosphorus, nitrogen and other components in the environment, while biotic can be further divided into producers (self nourishing e.g plants), consumers (things that consume producers to sustain life, e.g. animals, fish, insect, birds, mammals etc) and decomposers – bacteria and fungi that break down complex compounds from waste – dead producers and consumers – to make the chemical components available to producers] which can permanently upset the balance state of the natural cycles. – once this happens, then we have **Pollution** and the environment starts to impact on man.

Thermal



Air



Water



POLLUTION



Light



Visual



Noise



Land

Environmental Quotes

"Technology is a queer thing, it brings you great gifts with one hand, and stabs you in the back with the other" - C. P. Snow

"We do not inherit the earth from our ancestors; we borrow it from our children" - Chief Seattle

"We won't have a society if we destroy our environment"

- Margaret Mead

"When the last tree has been cut down
the last river poisoned
the last fish caught
and all the air unsafe to breathe,
then only will man discover
that he can not eat money"

- Cree (native American) Indian proverb

The Lady and the tiger

A king offered a challenge to three young men

his daughter, most desirable and suitable
a hungry tiger, fiercest and most cruel

Men	Action taken by man	Moral of the story
1 st	Refused the challenge- lived safe, died chaste	Can manage and survive without industries, the benefits and the problems they create
2 nd	Hired consultants, after a lot of money and time the consultants recommended a course of action – he opened the optimal door and was eaten by a low probability tiger	Do everything possible to prevent environmental problems
3 rd	Took a course in tiger handling	Take CHG 519 course and learn to deal with the situation-learn what has to be done to eliminate or minimize environmental problems

Note that the 3 men represent all of us, the lady the benefits industrialization brings to humanity, and the tiger the environmental problems associated with industrialization.

Welcome Aboard CHG 569